DOUBLE AUTHENTICATION SYSTEM INTEGRATING FACE RECOGNITION AND EYE BLINK COUNT RECOGNITION

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BONAFIDE CERTIFICATE

This is to certify that the project work entitled "**Double Authentication System Integrating Face Recognition and Eye Blink Count Recognition**" is a bonafide work done by **T.MADHAN KUMAR(18KB1A05E5), V.RAJASEKHAR(18KB1A05G0), BL.DHARMAJA(19KB5A0511), M.CHANDRIKA(19KB5A0514)** in the department of **Computer Science & Engineering, N.B.K.R.Institute of Science & Technology, Vidyanagar** and is submitted to **JNTUA, Ananthapuramu** in the partial fulfillment for the award of B.Tech degree in **Computer Science &Engineering.** This work has been carried out under my supervision.

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ABSTRACT

Authentication plays a major role in maintaining security in data accessing. It is the process of recognizing a user's identity and providing data access. It is a mechanism of associating an incoming request with a set of identifying credentials and providing access to data.

Traditional password authentication could not provide enough security. Even though many other authentication methods were introduced but unfortunately they couldn't prevent all types of security breaches, because of traces like finger prints, touch marks etc., were left during authentication.

Hence we come up with an idea named **"**Double Authentication System Integrating Face Recognition and Eye Blink Count Recognition**"**. This system provides double layered authentication for a user, that makes authentication more secure. The two layers of authentication are face recognition followed by eye blink count recognition. This system first verifies the user's face and then takes the eye blink count from the user and validate the credentials and provides an opening wedge for accessing data if the input credentials are valid.

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Biometric authentication is a security process that relies on the unique biological characteristics of individuals to verify them who they are. Biometric authentication systems compare physical or behavioural traits to store the authentic data in a database. If both samples of the biometric data matches, authentication is confirmed. Typically, biometric authentication is used to manage access to physical and digital resources, such as buildings, rooms and computing devices.

Biometric identification uses biometric, such as fingerprints or retina scans, to identify a person, whereas biometric authentication is the use of biometrics to verify people who they claim to be.

Face is one of the most important biometric features of a human. A human can recognize different faces without difficulty. Yet it is a challenging task to design a robust computer system for face identification. The inadequacy of automated face recognition systems is especially apparent when compared to our own innate face recognition ability. Human perform face recognition, an extremely complex visual task, almost instantaneously and our own recognition ability is far more robust than any computer's can hope to be. Human can recognize a familiar individual under very adverse lighting conditions, from varying angles or viewpoints. While research into this area dates back to the 1960's, it is only very recently the acceptable results have been obtained. However, face recognition is still an area of active research since a completely successful approach or model has not been proposed to solve the face recognition problem. The next generation surveillance systems are expected to take human face as input pattern and extract useful information such as face recognition from it.

1.2 MOTIVATION

The motivation for doing the project titled **“**Double Authentication System Integrating Face Recognition and Eye Blink Count Recognition**”** is to introduce a new authentication system which was not been earlier and provides more authenticity and security than existing authentication systems by overcoming their demerits.

The motivation for building a new and more secure authentication system made an open wedge to build an authentication system that leaves zero traces.

The idea about double layered authentication is that it integrates two different authentication systems (i.e., Face Recognition & Eye Blink Count Recognition) to make authentication system more secure.

1.3 AIM AND OBJECTIVE

The main aim of this project is to construct an authentication system which will not leave any traces on authentication by integrating face recognition and eye blink count models which leads to a new type of authentication system by providing double layer authentication.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

Literature survey is the most important step in software development process. Before developing the tools it is necessary to determine the time factor, economy and company strength. Once these things are satisfied, then next steps are to determine which operating system and language can be used for developing the tool. Once the programmers start building the tool the programmers need lot of external support. This support can be obtained from senior programmers, from books or from websites. Before building the system the above considerations are taken into account for developing the proposed system.

2.2LITERATURE SURVEY

[1] Liang, L., Ai, H.: Summary on Face Detection Researches. Journal of Computers 25(5), 449–459 (2004)

In this paper, it studies two critical technical links of face detection and face recognition in the recognition of face, and conducts analyses and comparisons on some commonly used face detections and recognition algorithms. Through the collation of this information and the combination of related knowledge in the digital image processing, it proposes a face recognition algorithm based on the singular value decomposition. After preprocessing face images, by the use of the projection method, it obtains positions of five sense organs, and then extracts local feature values near five sense organs as the main features of faces by using the singular value decomposition. For different face images of the same individual, the matching degree of feature values will be very high.

[2] Zhao, M.: Study on Coding Algorithm of Wavelet-based Color Face Images. Master Thesis. Computer Academy of Sciences of Sichuan Normal University, Sichuan (2005)

A face recognition system is one of the biometric information processes, its applicability is easier and working range is larger than others, i.e.; fingerprint, iris scanning, signature, etc. A face recognition system is designed, implemented and tested at Atılım University, Mechatronics Engineering Department. The system uses a combination of techniques in two topics; face detection and recognition. The face detection is performed on live acquired images without any application field in mind. Processes utilized in the system are white balance correction, skin like region segmentation, facial feature extraction and face image extraction on a face candidate. Then a face classification method that uses FeedForward

Neural Network is integrated in the system. The system is tested with a database generated in the laboratory with 26 people. The tested system has acceptable performance to recognize faces within intended limits. System is also capable of detecting and recognizing multiple faces in live acquired images.

[3] Liang, L., Ai, H.: Face Detections Based on Multi-Template Matching. Journal of China's Image Graphics 44(10), 623–630 (2004)

Face recognition has become relevant in recent years because of its potential applications. The aim of this paper is to find out the relevant techniques which give not only better accuracy also the efficient speed. There are several techniques available for face detection which give much better accuracy but the execution speed is not efficient. In this paper, a normalized cross-correlation template matching technique is used to solve this problem. According to the proposed algorithm, first different facial parts are detected likes mouth, eyes, and nose. If any of the two facial parts are found successfully then the face can be detected. For matching the templates with the target image, the template rotates at a certain angle interval.

[4] Chen, M.: Studies on the Face Image Detection and Classification System. PhD Thesis. Computer Science and Engineering Department of Shanghai Jiaotong University, Shanghai (2003)

The reliability of face recognition system has the characteristics of fuzziness, randomness, and continuity. In order to measure it in unconstrained scenes, we find out and quantify key broad-sense and narrow-sense influencing factors of reliability on the basis of analyzing operation states for six dynamic face recognition systems in the practical use of six public security bureaus. In this article, we propose a novel evaluation method with True Positive Identification Rate in dynamic and M:N mode and create a novel evaluation model of system reliability with the improved Fuzzy Dynamic Bayesian Network. Subsequently, we infer to solve the fuzzy reliability state probabilities of the six systems with Netica and get two most important factors with the improved fuzzy C-means algorithm. We verify the model by comparing the evaluation results with actual achievements of these systems. Finally, we find several vulnerabilities in the system with the least reliability and put forward a few optimization strategies. The proposed method combines advantages of the improved fuzzy C-means model with those of the dynamic Bayesian network to evaluate the reliability of the dynamic face recognition systems, making the evaluation results more reasonable and realistic. It starts a new research of face recognition systems in unconstrained scenes and contributes to the research on face recognition performance evaluation and system reliability

analysis. Besides, the proposed method is of practical significance in improving the reliability of the systems in use.

[5] Lu, C.: Study on Several Problems of Automatic Face Recognition and the System Implementation. PhD Thesis. Computer Academy of Sciences of Tsinghua University, Beijing (1998)

The paper analyses the multiple kernel learning-based face recognition in pattern matching area. Based on the analysis of the basic theory of multiple kernel SVM, this thesis focuses on the multiple kernel SVM algorithm based on semi-infinite linear program (SILP), including SILP based on column generation (CG) and SILP based on chunking algorithm (CA). The two SILP improved algorithms are applied to several classification problems, including UCI binary classification problem datasets and multi-classification problem datasets. Furthermore, the two SILP improved algorithms are applied to the actual problems of face recognition. The experiment data shows that with the multiple kernel learning-based method, the performance of face recognition can be obviously improved.

[6] Su, G.: Summary on Face Recognition Technology. Journal of China's Image Graphics 5(11), 220–238 (2000)

Face recognition presents a challenging problem in the field of image analysis and computer vision, and as such has received a great deal of attention over the last few years because of its many applications in various domains. Face recognition techniques can be broadly divided into three categories based on the face data acquisition methodology: methods that operate on intensity images; those that deal with video sequences; and those that require other sensory data such as 3D information or infra-red imagery. In this paper, an overview of some of the well-known methods in each of these categories is provided and some of the benefits and drawbacks of the schemes mentioned therein are examined. Furthermore, a discussion outlining the incentive for using face recognition, the applications of this technology, and some of the difficulties plaguing current systems with regard to this task has also been provided. This paper also mentions some of the most recent algorithms developed for this purpose and attempts to give an idea of the state of the art of face recognition technology.

[7] Li, J.: Research Progress of New Face Recognition Technology. Journal of Computers 31(10), 293–295 (2004)

Over the past few decades, interest in theories and algorithms for face recognition has been growing rapidly. Video surveillance, criminal identification, building access control, and unmanned and autonomous vehicles are just a few examples of concrete applications that are gaining attraction among industries. Various techniques are being developed including local,

holistic, and hybrid approaches, which provide a face image description using only a few face image features or the whole facial features. The main contribution of this survey is to review some well-known techniques for each approach and to give the taxonomy of their categories. In the paper, a detailed comparison between these techniques is exposed by listing the advantages and the disadvantages of their schemes in terms of robustness, accuracy, complexity, and discrimination. One interesting feature mentioned in the paper is about the database used for face recognition. An overview of the most commonly used databases, including those of supervised and unsupervised learning, is given. Numerical results of the most interesting techniques are given along with the context of experiments and challenges handled by these techniques. Finally, a solid discussion is given in the paper about future directions in terms of techniques to be used for face recognition.

2.3 EXISTING SYSTEM

There were many existing authentication systems like:

- Traditional password authentication
- Fingerprint authentication
- Face authentication
- Voice authentication, etc..

Every new authentication system previously introduced would try to overcome the cons of existing authentication systems.

But these existing authentication systems will leave traces of credentials while performing authentication and can be easily prone to security attack and security breaching.

In very first days of digital authentication era, the **traditional password authentication system** which uses a personal identification number (PIN) was used, but it still has demerits (leaves touch traces) that prone to unauthorized access.

To overcome this, biometric based authentication systems like **fingerprint authentication**, **face recognition**, **voice recognition** were introduced. But the fingerprint authentication system also leave the traces of thumb impression, and the face recognition can be cheated by using face picture of the user, and the voice recognition can be cheated by using recorded voice of user or through mimicry.

So, as every authentication system was introduced by overcoming some demerits of existing authentication systems, but still there are some traces left that are leading to unauthorized access to data.

But there is one authentication system named **Iris recognition authentication system** that provides fully secured authentication. As this system will recognize the iris of a person which is unique and cannot be faked. But this authentication system is expensive and cannot be affordable by all.

DEMERITS OF EXISTING SYSTEM

- The **traditional authentication** which uses personal identification number (PIN) for login, leaves the touch marks on the keypad, that can be used to trace out the PIN.
- The **fingerprint authentication** leaves the thumb impression of the user on the sensor, now a days that can be easily traced out and easily perform unauthorized access.
- The **face recognition** authentication system can be easily cheated by using face picture of the user.
- The **voice recognition** authentication system also be easily cheated by recorded audio of user or through mimicry.

2.4 PROPOSED SYSTEM

The proposed system **Double Authentication System Integrating Face Recognition and Eye Blink Count Recognition** will provide a double layered authentication for a user which makes the authentication more secure.

The proposed system was built by integrating face recognition model and eye blink count recognition model.

The flow of authentication is, firstly the face of the user was recognized by using face recognition model and then the password will be taken from user through eye blinks, this part will be done by the eye blink count recognition model.

For recognizing the face of user we need to build the face recognition model, which requires **Opencv** package and **HaarCascade** xml file and in the aspect of Machine Learning, we use a Classification Technique i.e., **Logistic Regression**(sigmoid). We train our model with the sample gray scale converted images of admin/user. Then we test our model with random faces of admin and checks the accuracy. If got good accuracy our face recognition model is ready to use.

For recognizing the eye blink count of a user we calculate the **Euclidean distance** between upper lid and lower lid of an eye of a user. The Euclidean distance becomes zero if the user closes the both lids and becomes some positive value if he opens the lids of an eye. The pair of consecutive one close and one open of lids is considered as a blink. In order to calculate the

Euclidean distance of eye lids of an user, we should be able to recognize the landmarks(which includes eyes) of a face.

Finally after collecting the face and the numeric password from the user, if the face and the password matches with the registered credentials, then access will be given to user to access data.

MERITS OF PROPOSED SYSTEM

- The proposed system will not leave any traces.
- Provides more security than existing system.
- Provides double layered authentication.
- Cost effective.
- Easily deployable.

2.5 FEASIBILITY ANALYSIS

The feasibility of the project is analysed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are:

- Economical Feasibility
- Technical Feasibility
- Social Feasibility

Economical feasibility

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

Technical feasibility

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

Social feasibility

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

CHAPTER 3

METHODOLOGY

3.1 METHODOLOGY

The proposed system was built by integrating face recognition model and eye blink count recognition model.

The flow of authentication is, firstly the face of the user was recognized by using face recognition model and then the password will be taken from user through eye blinks, this part will be done by the eye blink count recognition model.

3.1.1 MODULE 1 (Face Detection)

There are different packages used to detect face of user and they are mentioned below:

- cv2
- FaceMeshDetector

cv2

OpenCV is a great tool for image processing and performing computer vision tasks. It is an open-source library that can be used to perform tasks like face detection, objection tracking, landmark detection, and much more. It supports multiple languages including python.

In our project cv2 is mainly used for image processing. By using cv2 we convert the images into grayscale images which are very helpful in training the model.

FaceMeshDetector

Face Mesh Detection can be done using Python OpenCV directly via an Image file, webcam, or video file. This package is used to detect different parts on the face.

First of all a model should be created and that can be done by using LBPH algorithm.

LBPH Algorithm

Local Binary Pattern Histograms (LBPS) is a simple yet very efficient texture operator which labels the pixels of an image by thresholding the neighbourhood of each pixel and considers the result as a binary number.

It was first described in 1994 (LBP) and has since been found to be a powerful feature for texture classification. It has further been determined that when LBP is combined with histograms of oriented gradients (HOG) descriptor, it improves the detection performance considerably on some datasets.

Using the LBP combined with histograms we can represent the face images with a simple data vector.

As LBP is a visual descriptor it can be used for face recognition , as can be seen in the following step-by-step explanation.

1.Parameters

the LBPH uses 4 parameters:

- **Radius**: the radius is used to build the circular local binary pattern and represents the radius around the central pixel. It is usually set to 1.
- **Neighbours**: the number of sample points to build the circular local binary pattern. Keep in mind: the more sample points you include, the higher the computational cost. It is usually set to 8.
- **Grid X**: the number of cells in the horizontal direction. The more cells, the finer the grid, the higher the dimensionality of the resulting feature vector. It is usually set to 8.
- **Grid Y**: the number of cells in the vertical direction. The more cells, the finer the grid, the higher the dimensionality of the resulting feature vector. It is usually set to 8.

2. Training the Algorithm

First, we need to train the algorithm. To do so, we need to use a dataset with the facial images of the people we want to recognize. We need to also set an ID (it may be a number or the name of the person) for each image, so the algorithm will use this

information to recognize an input image and give you an output. Images of the same person must have the same ID.

With the training set already constructed, let's see the LBPH computational steps.

3. Applying the LBP operation

The first computational step of the LBPH is to create an intermediate image that describes the original image in a better way, by highlighting the facial characteristics. To do so,

the algorithm uses a concept of a sliding window, based on the parameters radius and neighbours.

The following image will show the procedure:

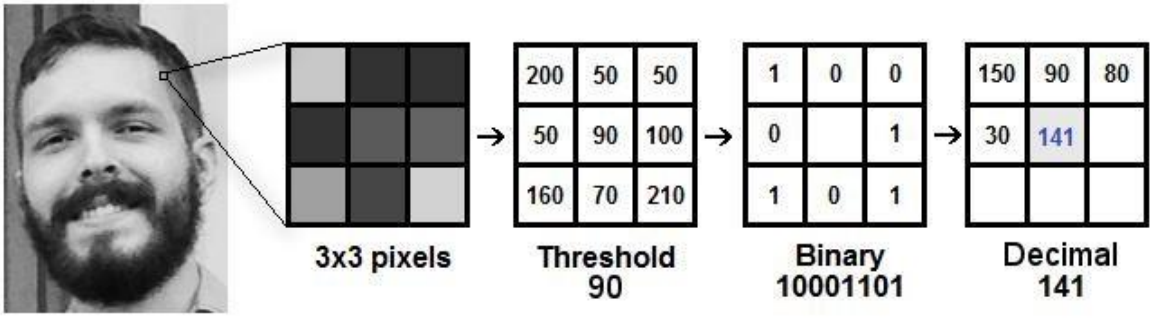


Fig: 3.1 procedure for dividing a pixel

- Suppose we have a facial image in grayscale.
- We can get part of this image as a window of 3x3 pixels.
- It can also be represented as a 3x3 matrix containing the intensity of each pixel (0~255).
- Then, we need to take the central value of the matrix to be used as the threshold.
- This value will be used to define the new values from the 8 neighbours.
- For each neighbour of the central value (threshold), we set a new binary value. We set 1 for values equal or higher than the threshold and 0 for values lower than the threshold.
- Now, the matrix will contain only binary values (ignoring the central value). We need to concatenate each binary value from each position from the matrix line by line into a new binary value (e.g. 10001101). Note: some authors use other approaches to concatenate the binary values (e.g. clockwise direction), but the final result will be the same.

- Then, we convert this binary value to a decimal value and set it to the central value of the matrix, which is actually a pixel from the original image.
- At the end of this procedure (LBP procedure), we have a new image which represents better the characteristics of the original image.
- **Note**: The LBP procedure was expanded to use a different number of radius and neighbours, it is called Circular LBP.

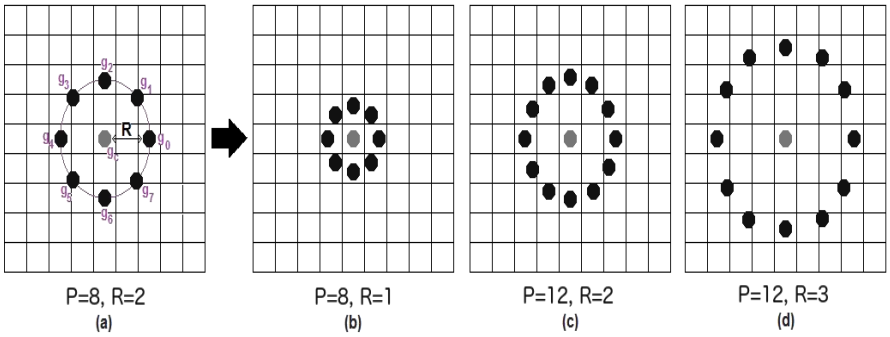


Fig: 3.2 Finding neighbours and radius

It can be done by using **bilinear interpolation**. If some data point is between the pixels, it uses the values from the 4 nearest pixels (2x2) to estimate the value of the new data point.

4. Extracting the Histograms

Now, using the image generated in the last step, we can use the **Grid X** and **Grid Y** parameters to divide the image into multiple grids, as can be seen in the following image:

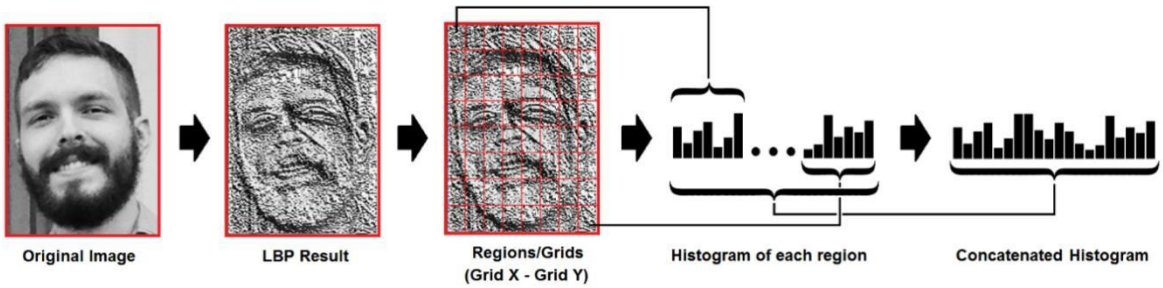


Fig: 3.3 Grayscale Conversion

Based on the image above, we can extract the histogram of each region as follows:

- As we have an image in grayscale, each histogram (from each grid) will contain only 256 positions (0~255) representing the occurrences of each pixel intensity.

- Then, we need to concatenate each histogram to create a new and bigger histogram. Supposing we have 8x8 grids, we will have 8x8x256=16.384 positions in the final histogram. The final histogram represents the characteristics of the image original image.

5. Performing the face recognition

In this step, the algorithm is already trained. Each histogram created is used to represent each image from the training dataset. So, given an input image, we perform the steps again for this new image and creates a histogram which represents the image.

- So to find the image that matches the input image we just need to compare two histograms and return the image with the closest histogram.
- We can use various approaches to compare the histograms (calculate the distance between two histograms), for example: **euclidean distance**, **chi-square**, **absolute value**, etc. In this example, we can use the Euclidean distance (which is quite known) based on the following formula:

$$D = \sqrt{\sum_{i=1}^{n} (hist1_i - hist2_i)^2}$$

Formula: 3.1 Euclidean Distance

- So the algorithm output is the ID from the image with the closest histogram. The algorithm should also return the calculated distance, which can be used as a '**confidence**' measurement.

Note: don't be fooled about the 'confidence' name, as lower confidences are better because it means the distance between the two histograms is closer.

- We can then use a threshold and the 'confidence' to automatically estimate if the algorithm has correctly recognized the image. We can assume that the algorithm has successfully recognized if the confidence is lower than the threshold defined.

After training done its time to recognize the face so we call a function named **face_check().** In this function we compare the trained model results and present input image. Then with the help of a mathematical formula we calculate the percentage of confidence. If the user is valid or not. If the confidence is more than 85% the user is valid and return "1" else user is not valid then function return "0".

3.1.2 Module 2 (Eye Blink Count Recognition)

For recognizing the eye blink count of a user we calculate the **Euclidean distance** between upper lid and lower lid of an eye of a user with the help of an user defined function named **blink()**. The Euclidean distance becomes zero if the user closes the both lids and becomes some positive value if he opens the lids of an eye.

The Euclidean distance is calculated with the help of pre-defined land marks on face, which was developed by Python for locating different points on the face. We will consider a list of points to locate eye position accurately and those list of points are given below

[22, 23, 24, 26, 110, 157, 158, 159, 160, 161, 130, 243]

The pair of consecutive one close and one open of lids is considered as a blink. In order to calculate the Euclidean distance of eye lids of user, we should be able to recognize the landmarks (which includes eyes) of a face. So we are using above given points as landmarks and we are taking horizontal and vertical distance between the eyelids and then calculate the average between the resultant values using the below formula

(Vertical length / Horizontal length) * 100

Formula: 3.2 Average

Now when the user's face was identified as authorized user, then only the system starts taking the numeric password through eye blink count. Here user need to give some time delay between the numbers and when some gap given by the user between the eye blinks, system consider the eyeblink count as a digit and appends to the password string. For example, the password is 123, at first user should blink eye for one time and wait for some time without any additional blinks and after that user need to give the 2 eyeblinks without any gap and it is continued for any length of password. After taking password from user our system will check the password and if password is correct, access to the database will be provided. If not a "wrong credentials" message will be displayed.

3.2 SYSTEM ARCHITECTURE

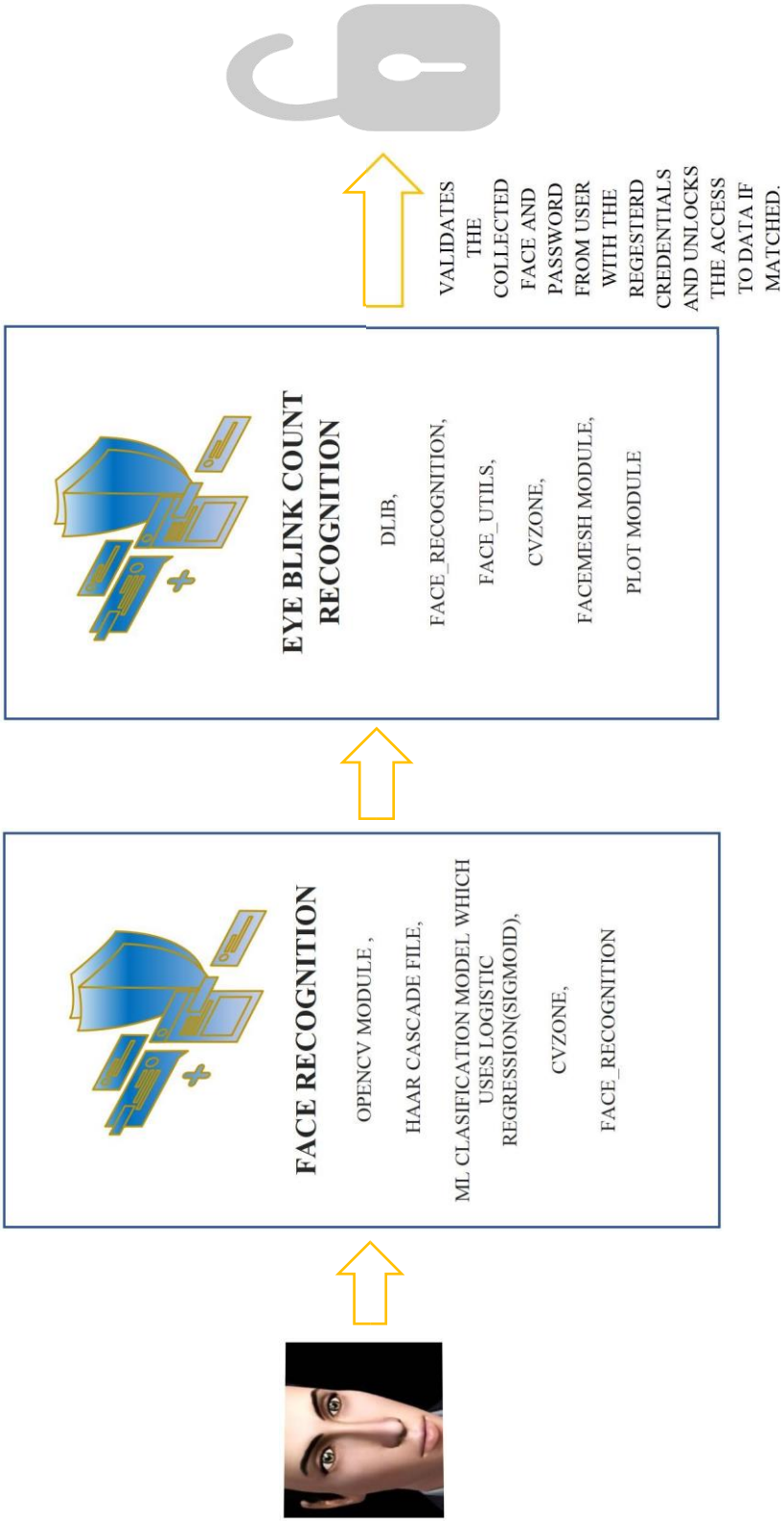


Fig-3.1

CHAPTER 4

IMPLEMENTATION ISSUES

4.1 SYSTEM REQUIREMENTS

4.1.1 SOFTWARE REQUIREMENTS

The software requirements are description of features and functionalities of the target system. Requirements convey the expectations of users from the software product. The requirements can be obvious or hidden, known or unknown, expected or unexpected from client's point of view.

- Pycharm IDE, Python-Flask Framework
- Libraries – cv2, dlib, imutils, scipy
- Database – MySQL

4.1.2 HARDWARE REQUIREMENTS

The hardware requirements are the requirements of a hardware device. Most hardware only has operating system requirements or compatibility. For example, a printer may be compatible with Windows XP but not compatible with newer versions of Windows like Windows 10, Linux, or the Apple macOS.

If a hardware device is not compatible with your computer, it is up to the manufacturer to release drivers. Unfortunately, many manufacturers only release updated drivers to fix problems with older drivers and often do not release drivers for newer operating systems or alternative operating systems. If a hardware device doesn't have drivers for your operating system, the only solution may be to get a more up-to-date replacement device.

- 64 bit laptop of 4gb RAM with running camera

4.2 DEPLOYMENT

After building a model we need to deploy that, here we are using our model for logging into a website. Hence we build a website for accessing data and it is done by using Flask framework.

Flask:

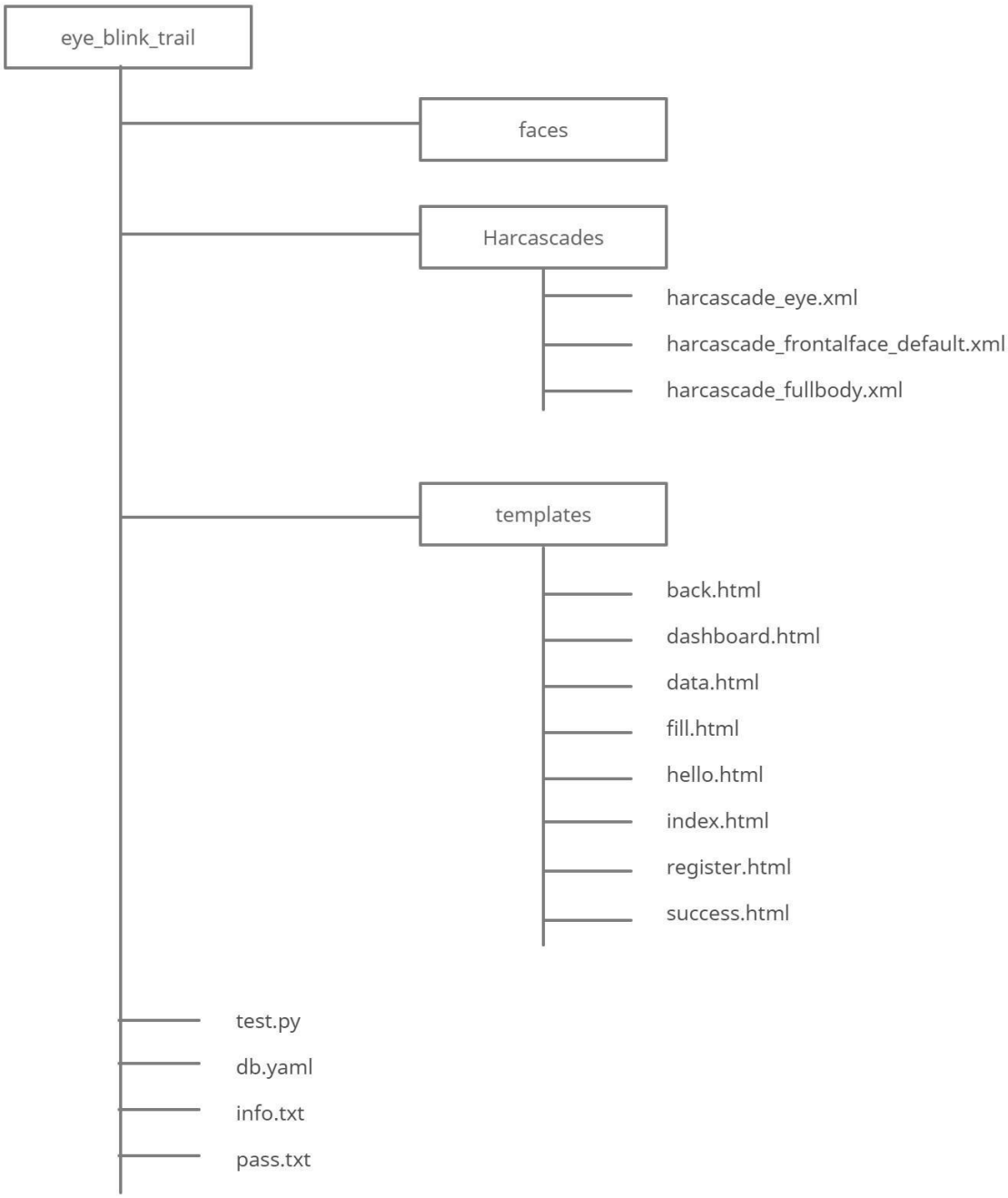
Flask is a web application framework written in Python. It was developed by Armin Ronacher, who led a team of international Python enthusiasts called Poocco. Flask is based on the Werkzeg WSGI toolkit and the Jinja2 template engine.Both are Pocco projects.

Unlike the Django framework, Flask is very Pythonic. It's easy to get started with Flask, because it doesn't have a huge learning curve.

On top of that it's very explicit, which increases readability. To create the "Hello World" app, you only need a few lines of code.

Flask is one of the most popular web frameworks, meaning it's up-to-date and modern. You can easily extend it's functionality. You can scale it up for complex applications.

4.3 DIRECTORY STRUCTURE



Lines in the code is 237

System space used by project is 97.3MB

SOURCE CODE

```
from flask import Flask, render_template,request,redirect,url_for,flash
import cv2
import cvzone
from cvzone.FaceMeshModule import FaceMeshDetector
from cvzone.PlotModule import LivePlot
import time
from flask_mysqldb import MySQL
import yaml
from tabulate import tabulate
import numpy as np
from os import listdir
from os.path import isfile, join


app=Flask(__name__)
app.secret_key = "super secret key"
db = yaml.safe_load(open('db.yaml'))
app.config['MYSQL_HOST'] = db['mysql_host']
app.config['MYSQL_USER'] = db['mysql_user']
app.config['MYSQL_PASSWORD'] = db['mysql_password']
app.config['MYSQL_DB'] = db['mysql_db']

mysql = MySQL(app)


face_classifier =
cv2.CascadeClassifier('Haarcascades/haarcascade_frontalface_default.xml')

def face_extractor(img):

    gray = cv2.cvtColor(img,cv2.COLOR_BGR2GRAY)
    faces = face_classifier.detectMultiScale(gray, 1.3, 5)
    if faces is ():
      return None

    for (x,y,w,h) in faces:
        cropped_face = img[y:y+h, x:x+w]
    return cropped_face



def face_detector(img, size=0.5):

   # Convert image to grayscale\n",
    gray = cv2.cvtColor(img,cv2.COLOR_BGR2GRAY)
    faces = face_classifier.detectMultiScale(gray, 1.3, 5)
```

```
    if faces is ():
        return img, [
    for (x,y,w,h) in faces:
        cv2.rectangle(img,(x,y),(x+w,y+h),(0,255,255),2),
        roi = img[y:y+h, x:x+w]
        roi = cv2.resize(roi, (200, 200))
    return img, roi

def im_collector():
    cap = cv2.VideoCapture(0)
    count = 0
    while True:
        ret, frame = cap.read()
        if face_extractor(frame) is not None:
            count += 1
            face = cv2.resize(face_extractor(frame), (200, 200))
            face = cv2.cvtColor(face, cv2.COLOR_BGR2GRAY)
            file_name_path = ''./faces/'' + str(count) + ''.jpg''
            cv2.imwrite(file_name_path, face)
            cv2.putText(face, str(count), (50, 50), cv2.FONT_HERSHEY_COMPLEX, 1, (0,
255, 0), 2)
            cv2.imshow(''Face Cropper'', face)
        else:
            print(''Face not found'')
            pass
        if cv2.waitKey(1) == 13 or count == 20:
            break
    cap.release()
    cv2.destroyAllWindows()


def train_data():
    data_path = ''./faces/''
    onlyfiles = [f for f in listdir(data_path) if isfile(join(data_path, f))]
    Training_Data, Labels = [], []
    for i, files in enumerate(onlyfiles):
        image_path = data_path + onlyfiles[i]
        images = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)
        Training_Data.append(np.asarray(images, dtype=np.uint8))
        Labels.append(i)
    Labels = np.asarray(Labels, dtype=np.int32)
    model = cv2.face.LBPHFaceRecognizer_create()
    model.train(np.asarray(Training_Data), np.asarray(Labels))
    return model


def face_check():
    model = train_data()
    cap = cv2.VideoCapture(0)
    while True:
```

```
        tm = time.time()
        ret, frame = cap.read()

        image, face = face_detector(frame)

        try:
            face = cv2.cvtColor(face, cv2.COLOR_BGR2GRAY)
            # Pass face to prediction model\n",
            # \"results\" comprises of a tuple containing the label and the confidence
value\n",
            results = model.predict(face)

            if results[1] < 500:
                confidence = int(100 * (1 - (results[1]) / 400))
                display_string = str(confidence) + '% Confident it is User'

            cv2.putText(image, display_string, (100, 120),
cv2.FONT_HERSHEY_COMPLEX, 1, (255, 120, 150), 2)

            if confidence >= 85:
                print(confidence)
                cv2.putText(image, "Unlocked", (250, 450),
cv2.FONT_HERSHEY_COMPLEX, 1, (0, 255, 0), 2)
                return 1
            else:
                cv2.putText(image, "Locked", (250, 450),
cv2.FONT_HERSHEY_COMPLEX, 1, (0, 0, 255), 2)
                cv2.imshow('Face Recognition', image)
                print(confidence)
                return 2


        except:
            cv2.putText(image, "No Face Found",\ (220, 120),
cv2.FONT_HERSHEY_COMPLEX, 1, (0, 0, 255), 2)
            cv2.putText(image, "Locked",\
 (250, 450), cv2.FONT_HERSHEY_COMPLEX, 1, \
(0, 0, 255), 2)
            cv2.imshow('Face Recognition', image)
            print('image not found')
            return 0

def blink():
    cap = cv2.VideoCapture(0)

    detector = FaceMeshDetector(maxFaces=1)
    plotY = LivePlot(640, 360, [20, 50], invert=True)
    idList = [22, 23, 24, 26, 110, 157, 158, 159,\ 160, 161, 130, 243]
    ratioList = []
    blinkCounter = 0
```

```
    counter = 0
    color = (255, 0, 255)
    delay_key = 0
    password1=open("pass.txt",'r')
    pass_word=password1.read()
    password1.close()
    fin_key = ''
    while True:
        tm = time.time()

        if cap.get(cv2.CAP_PROP_POS_FRAMES==\
cap.get(cv2.CAP_PROP_FRAME_COUNT):
            cap.set(cv2.CAP_PROP_POS_FRAMES, 0)

        success, img = cap.read()
        img, faces = detector.findFaceMesh(img, draw=False)

        if faces:
            face = faces[0]
            for id in idList:
                tuple(cv2.circle(img, face[id], 5, color, cv2.FILLED))
            leftUp = face[159]
            leftDown = face[23]
            leftLeft = face[130]
            leftRight = face[243]
            lenghtVer, _ = detector.findDistance(leftUp, leftDown)
            lenghtHor, _ = detector.findDistance(leftLeft, leftRight)
            cv2.line(img, leftUp, leftDown, (0, 200, 0), 3)
            cv2.line(img, leftLeft, leftRight, (0, 200, 0), 3)
            ratio = int((lenghtVer / lenghtHor) * 100)
            ratioList.append(ratio)
            if len(ratioList) > 3:
                ratioList.pop(0)
            ratioAvg = sum(ratioList)/\len(ratioList)
            if ratioAvg < 35 and counter == 0:
                blinkCounter += 1
                delay_key = 0
                color = (0, 200, 0)
                counter = 1
            if counter != 0:
                counter += 1
                if counter > 10:
                    counter = 0
                    color = (0, 0, 255)

            cvzone.putTextRect(img, f'Blink Count:\ {blinkCounter}', (50, 100),
                        colorR=color)
            imgPlot = plotY.update(ratioAvg, color)
            img = cv2.resize(img, (640, 360))
            imgStack = cvzone.stackImages([img\ ,imgPlot], 2, 1)
```

```
        else:
            img = cv2.resize(img, (640, 360))
            imgStack = cvzone.stackImages([img,\ img], 2, 1)
        delay_key += 1647701038
        if delay_key == (30 * 1647701038):
            if len(fin_key) == 0 and blinkCounter\ == 0:
                pass
            else:
                fin_key += str(blinkCounter)
            blinkCounter = 0
        print(fin_key)
        if len(fin_key) == len(pass_word):
            break
        cv2.imshow("Image", imgStack)
        cv2.waitKey(25)
    print(tm)
    return fin_key




@app.route('/')
def index():
    return render_template("index.html")


@app.route('/reset')
def reset():
    face_class=face_check()
    if face_class==1:
        password1 = open("pass.txt", 'r')
        pass_word = password1.read()
        password1.close()
        if blink() == pass_word:
            im_collector()
            train_data()
            k=blink()
            f=open("pass.txt","w")
            f.write(k)
            f.close()
    flash("oops! something went wrong try again!")
    return render_template('dashboard.html')
@app.route('/register')
def register():
    face_class=face_check()
    if face_class==1:
        password1 = open("pass.txt", 'r')
        pass_word = password1.read()
        password1.close()
        if blink() == pass_word:
```

```
            im_collector()
            train_data()
            k=blink()
            f=open("pass.txt","w")
            f.write(k)
            f.close()
    flash("oops! something went wrong try again!")
    return render_template('index.html')

@app.route('/auth')
def auth():
    var = face_check()
    if var==1:
        password1 = open("pass.txt", 'r')
        pass_word = password1.read()
        password1.close()
        if blink()==pass_word:
            return render_template("dashboard.html")
        else:
            return render_template('back.html')
    elif var == 0:
        return render_template('back.html')
    elif var == 2:
        return render_template('back.html')
    else:
        return render_template('back.html')


@app.route('/dash', methods=['GET', 'POST'])
def dash():
        return render_template('dashboard.html')




@app.route('/fill', methods=['GET', 'POST'])
def fill():

    if request.method == 'POST':
        # Fetch form data
        userDetails = request.form

        event = userDetails['event']
        dt = userDetails['dt']
        orga = userDetails['organizer']
        cur = mysql.connection.cursor()
        cur.execute("INSERT INTO events(event, dt\ ,organizer) VALUES(%s,
%s,%s)",(event, dt, orga))
        mysql.connection.commit()
```

```
        cur.close()
        return render_template('succes.html')
    return render_template('fill.html')



@app.route('/data')
def users():
    cur = mysql.connection.cursor()
    resultValue = cur.execute("SELECT * FROM events")
    if resultValue > 0:
        userDetails = cur.fetchall()
        with open("info.txt", "w") as outputfile:
            outputfile.write(tabulate(userDetails))

        return render_template('data.html',userDetails=userDetails)



if __name__=="__main__":
    app.run(debug=True)
```

CHAPTER 5

TESTING

5.1 TESTING

The following testings have been done for the project:

1.Unit Testing

2.Integration Testing

3.System Testing

5.1.1 UNIT TESTING

In this, each and every component has been tested to verify whether they are providing accurate results efficiently without precision errors.

5.1.2 INTEGRATION TESTING

Integration tests are designed to test integrated software components to determine if they actually run as one program and observed the result.

5.1.3 SYSTEM TESTING

System testing is testing conducted on a complete integrated system to evaluate the system's compliance with its specified.

5.2 TEST CASES

Testing if authentication is successful:

Test No:	Action	Inputs	Expected output	Actual output	Test Browser	Test Result	Test Comments
1	Launch Website	Local Host	Login page of the website	Login page of the website	Google Chrome	Pass	Launch Successful
2	Perform face authentication	User’s face	User’s face was authorized	User’s face was authorized	Google Chrome	Pass	Face authentication was successful

3	convey the registered password through eye blinks	Password through eye blinks	Password matched and navigates to home page	Password matched and navigates to home page	Google Chrome	Pass	Two levels of authentication was successful

Table: 5.1 Successful Authentication

Testing if authentication is failed:

Case1: If 1st level of authentication failed

Test No:	**Action**	**Inputs**	**Expected output**	**Actual output**	**Test Browser**	**Test Result**	**Test Comments**
1	Launch Website	Local Host	Login page of the website	Login page of the website	Google Chrome	Pass	Launch Successful
2	Perform face authentication	User's face	User's face was authorized	User's face was not authorized navigates to retry page	Google Chrome	Fail	Face authentication was not successful

Table: 5.1 1st Level Authentication Failure

Case1: If 2nd level of authentication failed

Test No:	**Action**	**Inputs**	**Expected output**	**Actual output**	**Test Browser**	**Test Result**	**Test Comments**
1	Launch Website	Local Host	Login page of the website	Login page of the website	Google Chrome	Pass	Launch Successful

2	Perform face authentication	User's face	User's face was not authorized	User's face was authorized	Google Chrome	Pass	Face authentication was successful
3	convey the registered password through eye blinks	Password through eye blinks	Password matched and navigates to home page	Password not matched and navigates to retry page	Google Chrome	Fail	2nd level of authentication was not successful

Table: 5.1 2 Levels Authentication Failure

5.3 RESULTS

Login page will looks like as following image:

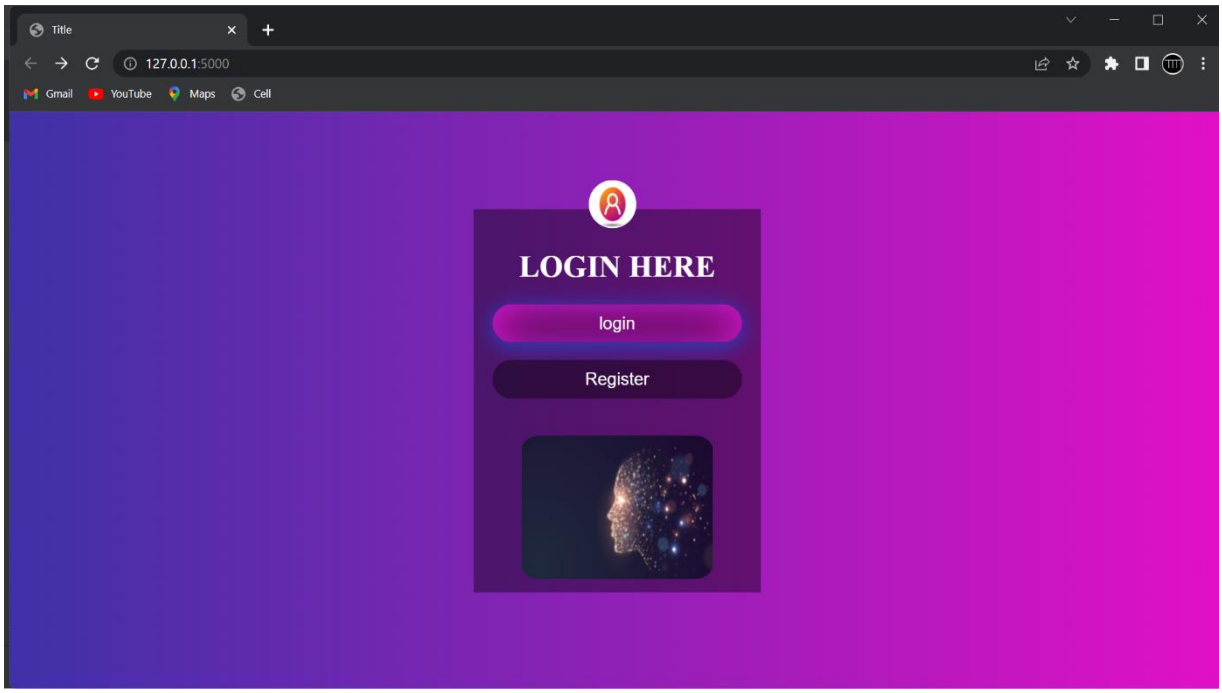


Fig: 5.1 Login page

Performing authentication looks like this:

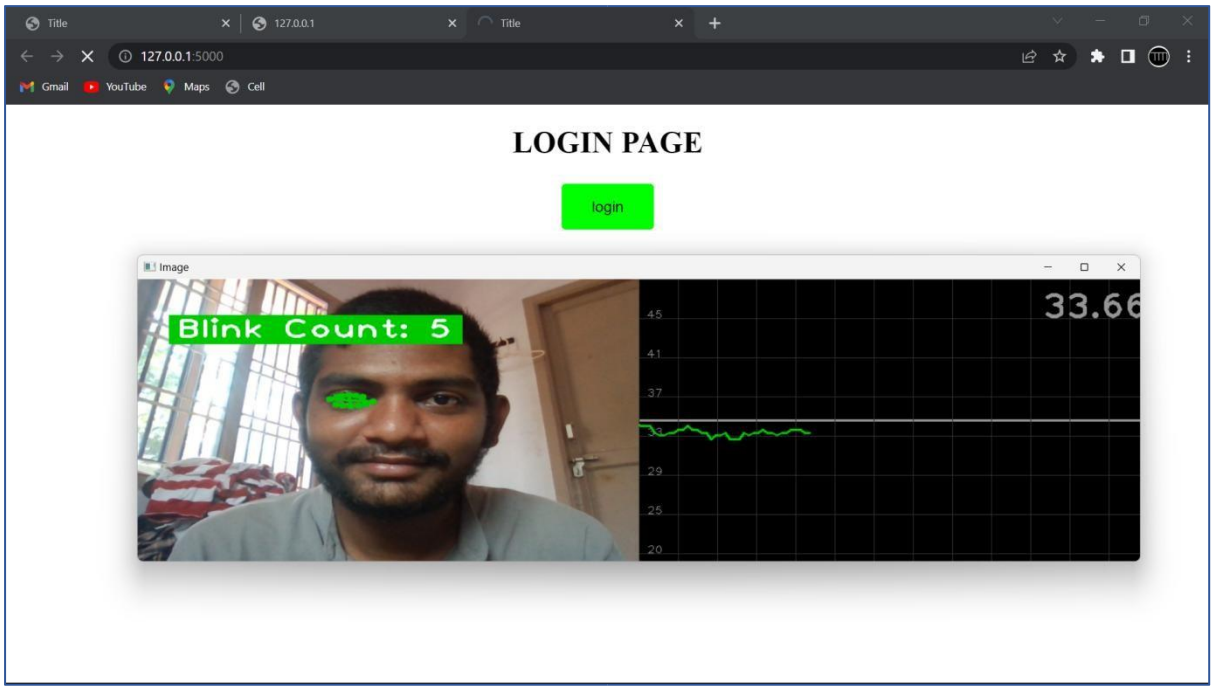


Fig: 5.2 Performing authentication

If wrong credential were given the following page will be displayed:

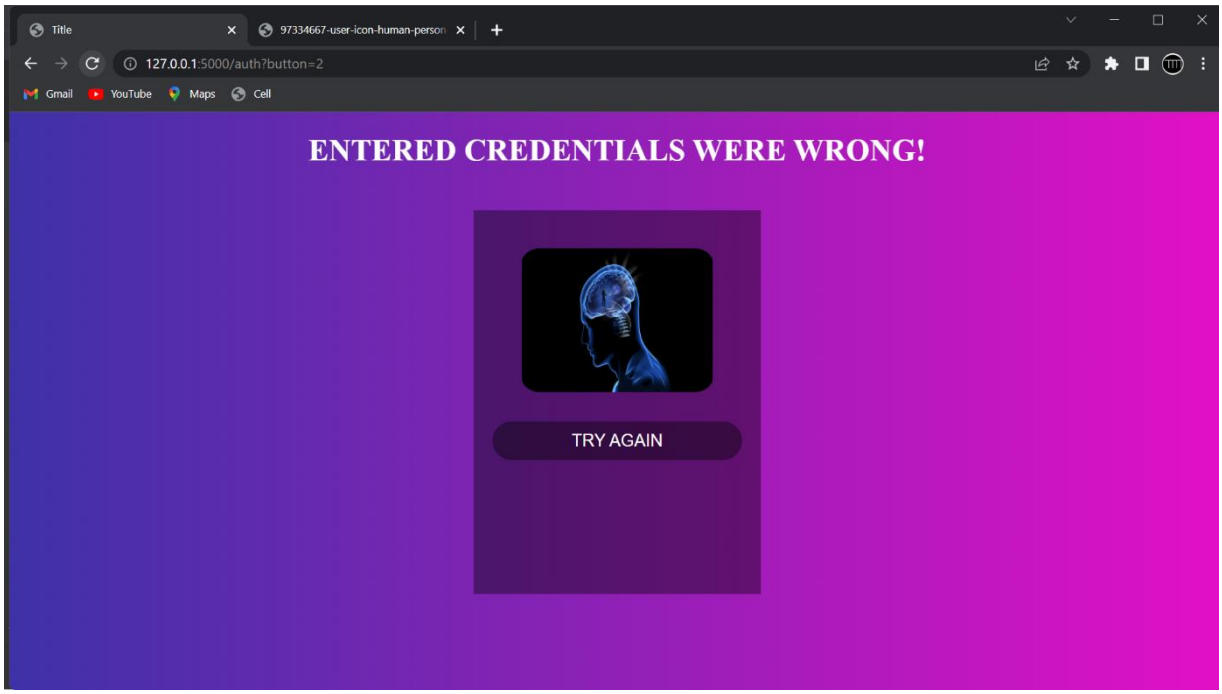


Fig: 5.3 Page for retrying the authentication process

If correct credentials were given the following page will be displayed:

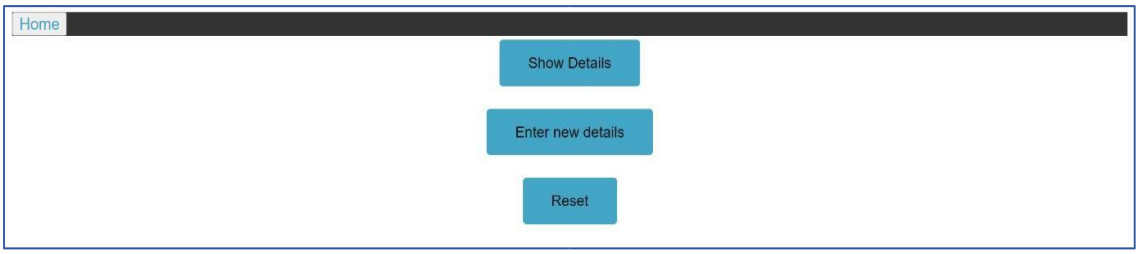


Fig: 5.4 Dashboard

From this page data will be accessed.

5.4 RESULT COMPARISION WITH EXISTING SYSTEM

- **Publication of faking Fingerprint authentication system:**

1.Using a neural network, researchers from New York University (NYU) evolved a fake fingerprint that could fool biometric authentication for one in five people.

NEW YORK: Scientists have developed an artificial intelligence tool that can synthesise fake human fingerprints and potentially fool biometric authentication systems.

Fingerprint authentication systems are a widely trusted, ubiquitous form of biometric authentication, deployed on billions of smartphones and other devices worldwide.

However, researchers from New York University (NYU) in the US revealed a surprising level of vulnerability in these systems.

Using a neural network, they evolved a fake fingerprint that could fool biometric authentication for one in five people.

Much the way that a master key can unlock every door in a building, these "DeepMasterPrints" use artificial intelligence to match a large number of prints stored in fingerprint databases and could thus theoretically unlock a large number of devices.

The work builds on earlier research led by Nasir Memon, a professor at NYU. Memon, who coined the term "MasterPrint," described how fingerprint-based systems use partial fingerprints, rather than full ones, to confirm identity.

Devices typically allow users to enroll several different finger images, and a match for any saved partial print is enough to confirm identity.

Partial fingerprints are less likely to be unique than full prints, and Memon's work demonstrated that enough similarities exist between partial prints to create MasterPrints capable of matching many stored partials in a database.

Researchers took this concept further, training a machine-learning algorithm to generate synthetic fingerprints as MasterPrints.

"Fingerprint-based authentication is still a strong way to protect a device or a system, but at this point, most systems don't verify whether a fingerprint or other biometric is coming from a real person or a replica," said doctoral student Philip Bontrager.

"These experiments demonstrate the need for multi-factor authentication and should be a wake-up call for device manufacturers about the potential for artificial fingerprint attacks," he said.

Reference

https://www.newindianexpress.com/lifestyle/tech/2018/nov/26/aigenerates-fake-fingerprints-to-fool-biometric-systems-1903434.html

2. The Tihar Prison Department has stopped salaries of 47 officials after a mismatch was detected in their biometric verification, an official said on Saturday.

Directed General (Prisons) Sandeep Goel confirmed the development.

A senior official said that biometric verification drive was conducted by the Delhi Subordinate Services Selection Board (DSSSB) in Prisons Department in the last week of November. “All the new candidates from 2019 onwards were screened by matching their biometrics with the data preserved at the time of recruitment,” the official said.

The result of the exercise revealed that preliminary mismatch have been detected in 47 cases. “Further action will be taken after the DSSSB final report,” the official said.

However, as an immediate and interim measure, the salaries of all these officials have been stopped and show-cause notices have been issued to them, the Prison Department said.

Reference

https://www.thehindu.com/news/cities/Delhi/biometric-fraud-hitstihar/article38039368.ece

- But our proposed Authentication system will not prone to those security breaches unlike existing systems which were reported in the above publications. As our proposed authentication system does not leave any traces like fingerprints, touch traces, etc unlike the existing authentication systems. So, our authentication system will make user to experience a secure authentication.

CHAPTER 6
CONCLUSION AND FUTURE ENHANCEMENTS

6.1 CONCLUSION

The proposed "Double Authentication System Integrating Face Recognition and Eye Blink Count Recognition" was a new authentication system that overcomes the demerits regarding security attacks and security breaches of existing authentication systems and methodologies by providing double authentication and leaving zero traces while performing authentication.

6.2 FUTURE ENHANCEMENTS

- Deploying the authentication system idea with IOT.
- Providing a feature that allows the other user to login through OTP with the permission of that authorized user during his/her physical absence.
- Providing a feature for blind users that gives voice notifications during the start and at the end of the login process.
    1. If the login is successful, it should say "logged in successfully",
    2. If the password was wrong, it should say "Invalid password",
    3. If the user face was unauthorized, it should say "Invalid face"
    4. If face was not visible, it should say "Face not found"

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- *Zhang, J., He, T.: Face Recognition Method Based on Extractions of Facial Geometric Feature Points. Infrared and Laser Engineering 28(4), 40–43 (1999)*

TEXT BOOKS

- Introduction to Machine Learning with Python: A Guide for Data Scientists, book by Andreas C.Mullar and Sarah Guido, Grayscale Indian Edition.
- Machine Learning Engineering, a book by Andriy Burkov, First Edition
- Learn Python Programming, book by Fabrizio Romano, Second Edition

e-references

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- https://realpython.com/face-recognition-with-python/
- https://www.youtube.com/watch?v=mI9Hjlp2cHs